KOGANTI SASANK

AI Security Engineer and Senior Data Scientist

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PROFESSIONAL SUMMARY

Senior Data Scientist with 4+ years of experience building production AI and Generative AI systems, with hands-on work securing RAG and multi-agent applications. Experience spans indirect prompt-injection controls, data-leakage prevention, secure document ingestion, role-aware retrieval, restricted tool access, guardrails, private LLM deployment, output validation, observability and auditability.

CORE SKILLS

AI application security: Prompt injection defense, secure RAG, data leakage prevention, input and output validation, AI guardrails, role-based access, human-in-the-loop controls
Secure ingestion: ClamAV malware scanning, file and MIME validation, content sanitization, document validation, PII and secret detection, controlled corpus inclusion
AI platforms: LangGraph, NeMo Guardrails, Elasticsearch, Langfuse, Ollama, vLLM, Llama, Gemma, hybrid retrieval
Engineering security: Python, FastAPI, SQL, REST APIs, Docker, Kubernetes, Bandit, pytest, audit logging, private model deployment

PROFESSIONAL EXPERIENCE

Senior Data Scientist | TCG Digital, Remote, India | Jul 2025 - Present

- Architected an enterprise multi-agent procurement platform with five specialized agents and a supervisor, incorporating security controls across ingestion, retrieval, agent execution, tool use and response generation.
- Hardened the RAG ingestion pipeline against indirect prompt injection by treating uploaded and external documents as untrusted content before inclusion in the retrieval corpus.
- Implemented ClamAV malware scanning, file and MIME validation, content sanitization and document validation before indexing content in Elasticsearch.
- Added prompt-injection and data-leakage checks for incoming documents, retrieved context and generated responses to reduce malicious instruction execution and confidential-data exposure.
- Designed role-aware retrieval, restricted tool access, structured inputs and outputs, validation boundaries, confidence checks and human-review paths for sensitive or uncertain actions.
- Applied NeMo Guardrails and validation layers for relevance, safety, information-leakage checks and controlled LLM interactions.
- Implemented structured logging and Langfuse observability for agent decisions, retrieved evidence, tool interactions and generated responses to support traceability and audit review.
- Used self-hosted Llama and Gemma models through Ollama and vLLM for data-sensitive workloads, reducing exposure of confidential enterprise information to external AI services.

Data Scientist Consultant | TCG Digital, Remote, India | Jul 2022 - Jun 2025

- Built document-intelligence pipelines for large volumes of PDFs and technical records with controlled parsing, filtering, relevance evaluation and corpus inclusion.
- Developed a patent-intelligence system processing 50,000+ documents with staged retrieval and LLM-based relevance assessment before downstream use.
- Built a private proposal-generation platform using self-hosted Llama 3.1 and vLLM to keep confidential documents inside controlled infrastructure.
- Created a reusable CI/CD framework with Bandit scans, pytest tests, golden-dataset checks and Jenkins deployment for FastAPI and Flask services.

SECURITY ENGINEERING APPROACH

Secure AI Lifecycle Controls

- Applies layered controls across document intake, retrieval, model execution, agent permissions, output validation, monitoring and audit rather than relying on a single prompt filter.

LEADERSHIP AND RECOGNITION

Contributes to production AI architecture, design reviews and engineering standards while mentoring colleagues and interviewing GenAI engineering candidates.

EDUCATION

M.Sc. Mathematics | BITS Pilani | 2018 - 2022